



Woodruff Scientific, Inc

Revisit of the 2017 Costing for Four ARPA-E ALPHA Concepts
DE-AR001175

Award:	DE-AR001175
Sponsoring Agency	USDOE, Advanced Research Project Agency – Energy (ARPA-E)
Lead Recipient:	Woodruff Scientific
Project Team Members	Lucid Catalyst, Decysive Systems
Program Director:	Dr. Scott Hsu
Principal Investigator:	Dr. Simon Woodruff
Tech SETA:	Dr. Colleen Nehl
Date of Report:	10/19/2020

	Average	Low	High	
20. Land/Rights	14.0	10.3	22.0	M\$
21. Structures/Site	234.7	173.9	369.7	M\$
22. Reactor Plant Equip.	422.7	268.8	557.3	M\$
22.1 Reactor Equip.	172.1	95.0	243.2	M\$
22.1.1 First Wall/Blanket	57.3	3.6	116.5	M\$
22.1.2 High Temp. Shield	24.9	22.7	28.5	M\$
22.1.3 Coils	5.9	0.0	22.8	M\$
22.1.4 Suppl. Heating	3.0	0.0	12.0	M\$
22.1.5 Primary Structure	11.6	7.1	15.8	M\$
22.1.6 Vacuum System	1.4	0.1	4.7	M\$
22.1.7 Power Supplies	55.8	11.9	140.4	M\$
22.1.8 Plasma Source	1.4	0.6	3.0	M\$
22.1.9 Direct E. Conv.	0.0	0.0	0.0	M\$
22.1.10 ECRH	0.0	0.0	0.0	M\$
22.1.11 Assembly and installation	10.8	0.3	36.5	M\$
22.2 Main Heat Transfer	113.2	63.6	184.2	M\$
22.3 Auxiliary Cooling	2.6	1.3	4.2	M\$
22.4 Rad. Waste Treat.	4.6	2.4	7.4	M\$
22.5 Fuel Processing	123.8	92.3	176.0	M\$
22.6 Other plant equipment	4.2	2.1	6.7	M\$
22.7 Instrumentation and control	2.1	0.0	3.9	M\$
23. Turbine Plant Equip.	137.5	101.9	216.6	M\$
24. Electric Plant Equip.	58.9	43.7	92.8	M\$
25. Misc. Plant Equip.	39.8	31.9	55.4	M\$
26. Heat Rejection	55.3	41.0	87.1	M\$
27. Special Materials	103.1	1.4	266.9	M\$
90. Total Direct Cost	1066.1	711.0	1454.7	M\$
91. Construction Serv./Mat.	26.1	19.2	33.7	M\$
92. Home Office Eng./Serv.	31.3	23.0	40.5	M\$
93. Field Office Eng./Serv.	10.4	7.7	13.5	M\$
94. Owners Cost	37.8	27.7	48.8	M\$
96. Contingency	0.0	0.0	0.0	M\$
97. Interest During Constr.	51.7	37.9	66.8	M\$
99. Total Capital Cost:	1223.5	837.7	1638.8	M\$

Table 3. Capital costs showing the average, low and high values.